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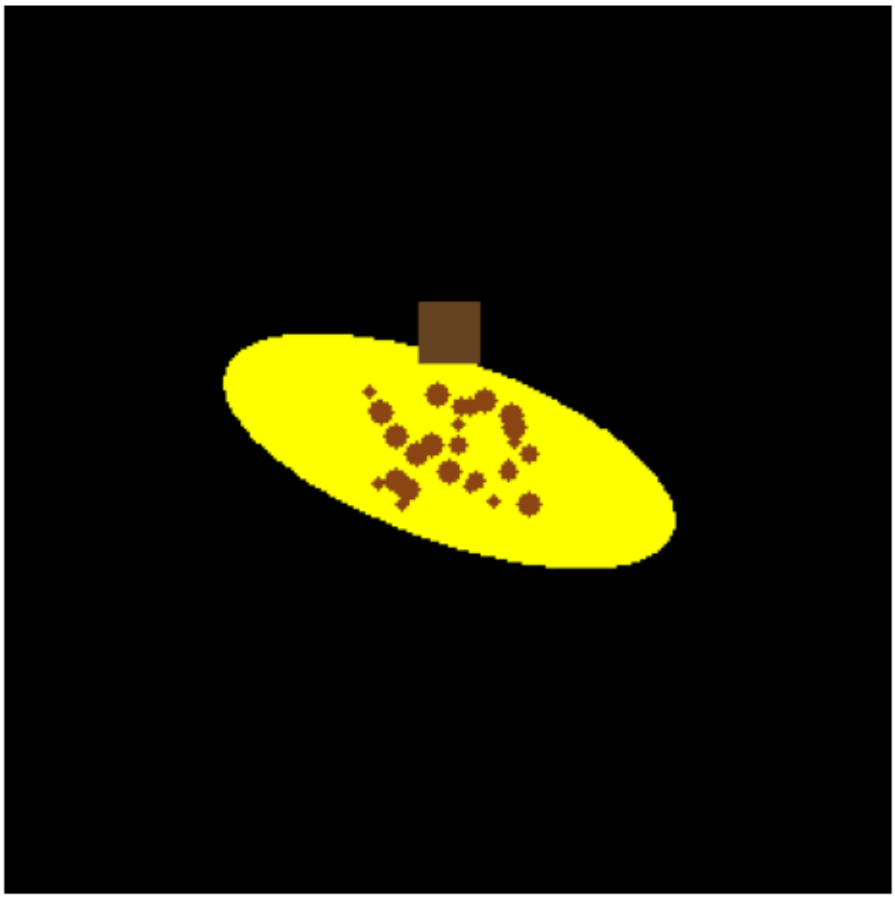
BANANA UNIT-3

TASK 1: Vanilla Gradient Saliency Map for Banana Classification

Synthetic Banana Image

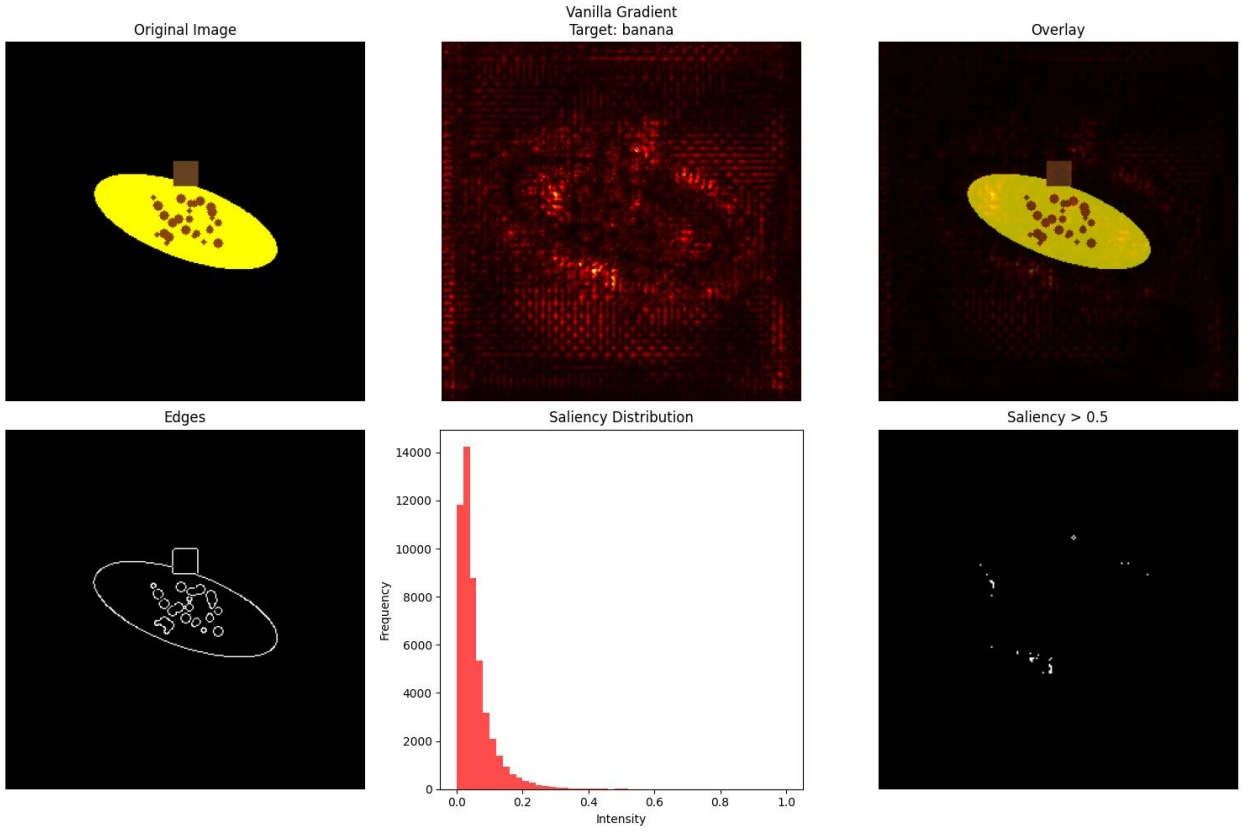


Predicted class: pick

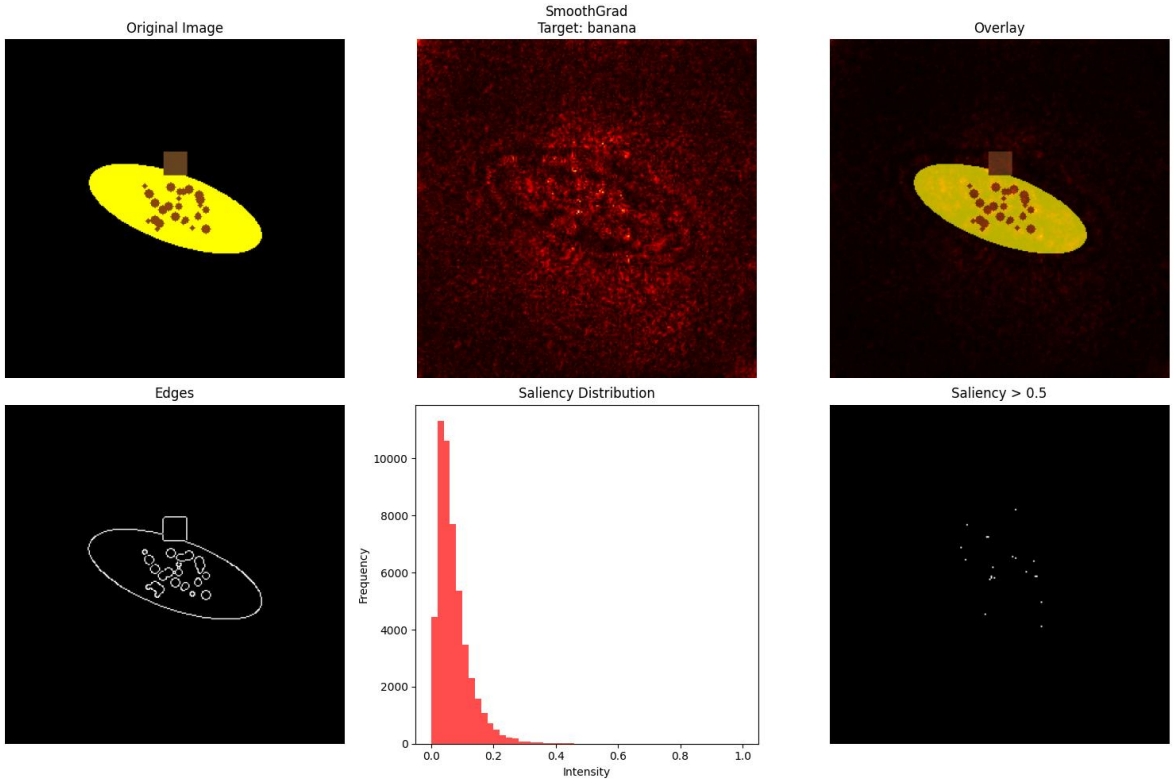
Confidence: 0.8298

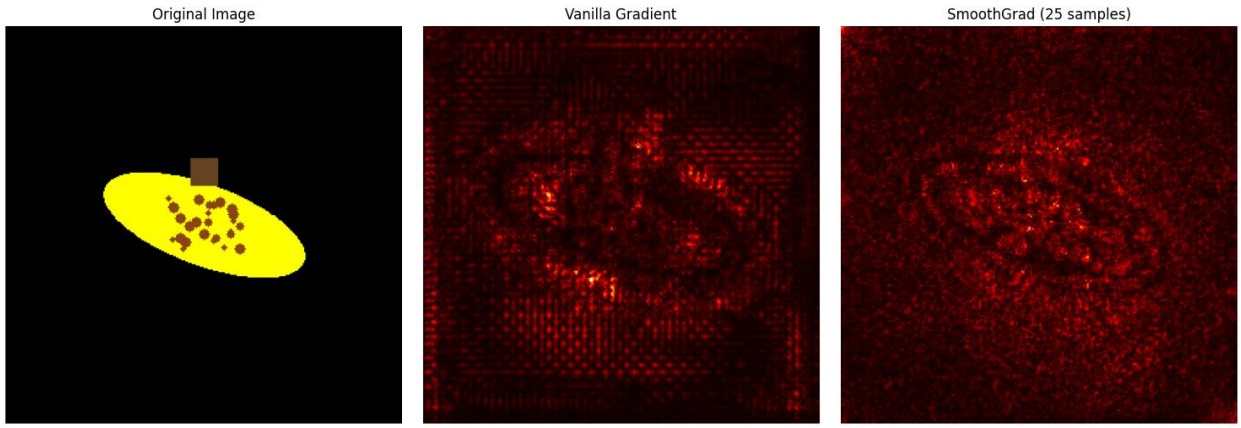
Banana class index: 954 (banana)

Generating vanilla gradient saliency map...



After computing SmoothGrad with 25 samples...





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INTERPRETATION FOR TASK 1:

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Vanilla Gradient Saliency Map Analysis:

1. Edge Highlighting:

✓ The saliency maps primarily highlight the edges and contours of the banana

✓ Strong activations appear along the boundaries where the banana meets the background

✓ The curved shape of the banana is clearly outlined

2. Noise Characteristics:

✓ Vanilla gradients appear noisy due to gradient saturation

✓ Many small, scattered activations are visible throughout the image

✓ SmoothGrad significantly reduces noise by averaging multiple samples

3. Background Interference:

✓ Background edges may also appear in the saliency map

✓ The synthetic image has minimal background, but edges of the banana itself are highlighted

✓ This demonstrates that vanilla gradients focus on high-frequency information (edges)

4. Object Region Activation:

✓ While edges dominate, some internal regions show moderate activation

✓ The activation is stronger on distinctive features (brown spots on the banana)

✓ Internal regions are less highlighted compared to edges

5. Vanilla vs SmoothGrad Comparison:

✓ Vanilla gradient: Noisier, sharper edge responses

✓ SmoothGrad: Cleaner, more continuous regions

✓ SmoothGrad better highlights the entire banana shape, not just edges

Key Observations:

- Vanilla gradients provide a good first approximation of important image regions
- The method is sensitive to edges and high-contrast boundaries
- SmoothGrad effectively reduces noise and produces more coherent saliency maps
- Both methods successfully identify the banana's location and shape

TASK 2:

TASK 2: Smooth Visualization via Activation Maximization

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CONFIGURATION 1: Basic (with regularization)

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Starting activation maximization for class 954

Optimization parameters:

- Iterations: 500
- Learning rate: 0.05
- L2 regularization: 0.001
- TV regularization: 0.001
- Gaussian blur: True (sigma=0.5)

Iteration 0: class score = 2.0254

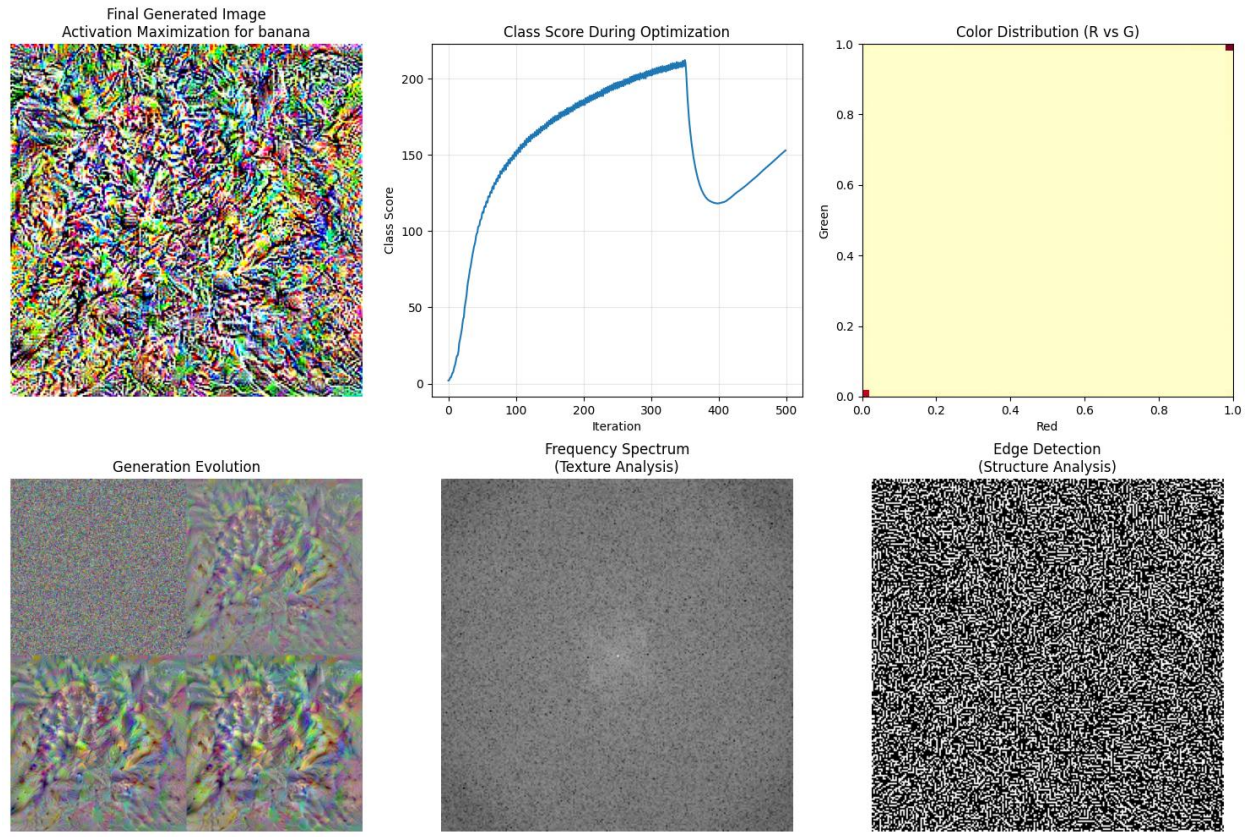
Iteration 100: class score = 149.1846

Iteration 200: class score = 186.0726

Iteration 300: class score = 204.2178

Iteration 400: class score = 118.2729

Final iteration 500: class score = 152.9109



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CONFIGURATION 2: Aggressive (stronger regularization)

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Starting activation maximization for class 954

Optimization parameters:

- Iterations: 500
- Learning rate: 0.03
- L2 regularization: 0.01
- TV regularization: 0.01
- Gaussian blur: True (sigma=1.0)

Iteration 0: class score = 1.9388

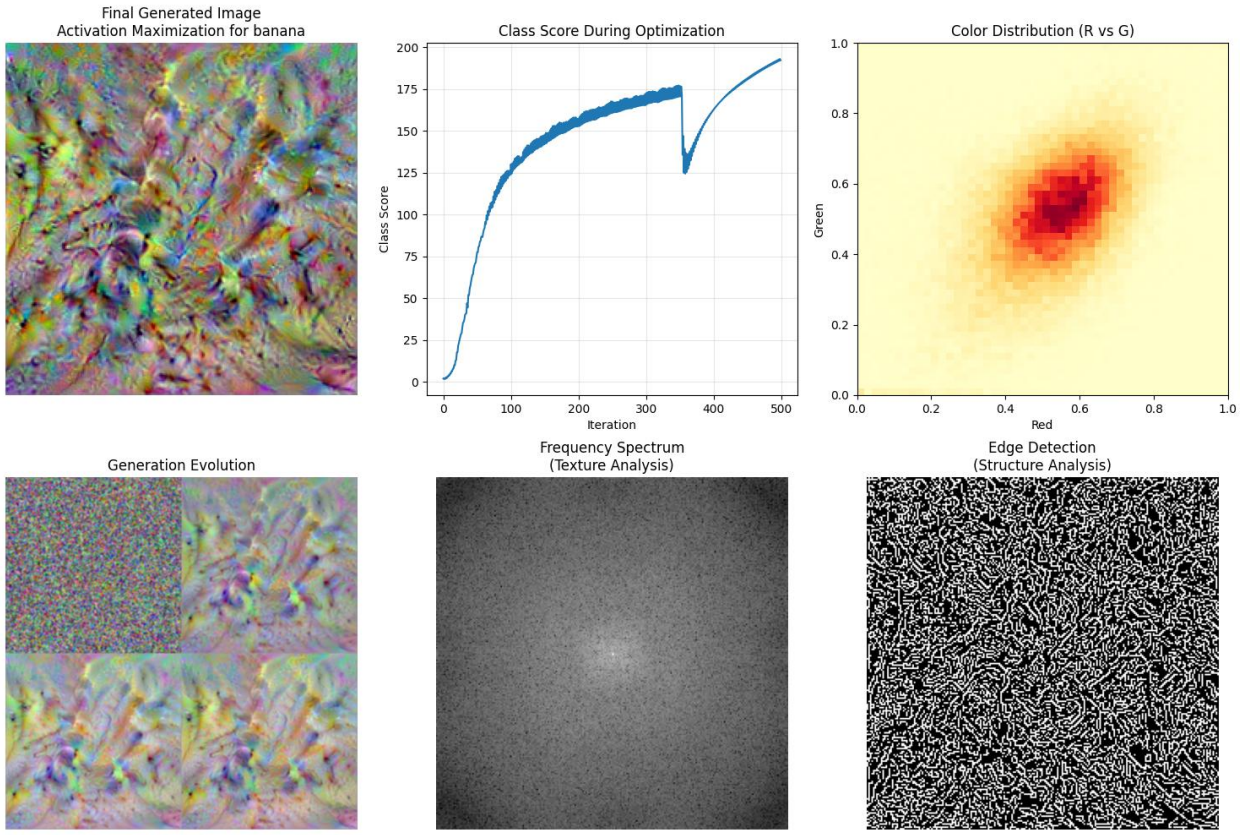
Iteration 100: class score = 130.1063

Iteration 200: class score = 157.4879

Iteration 300: class score = 172.5728

Iteration 400: class score = 162.6352

Final iteration 500: class score = 192.2917



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CONFIGURATION 3: Minimal regularization (more noise)

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Starting activation maximization for class 954

Optimization parameters:

- Iterations: 500
- Learning rate: 0.1
- L2 regularization: 0.0001
- TV regularization: 0.0001

- Gaussian blur: False (sigma=0.5)

Iteration 0: class score = 2.1484

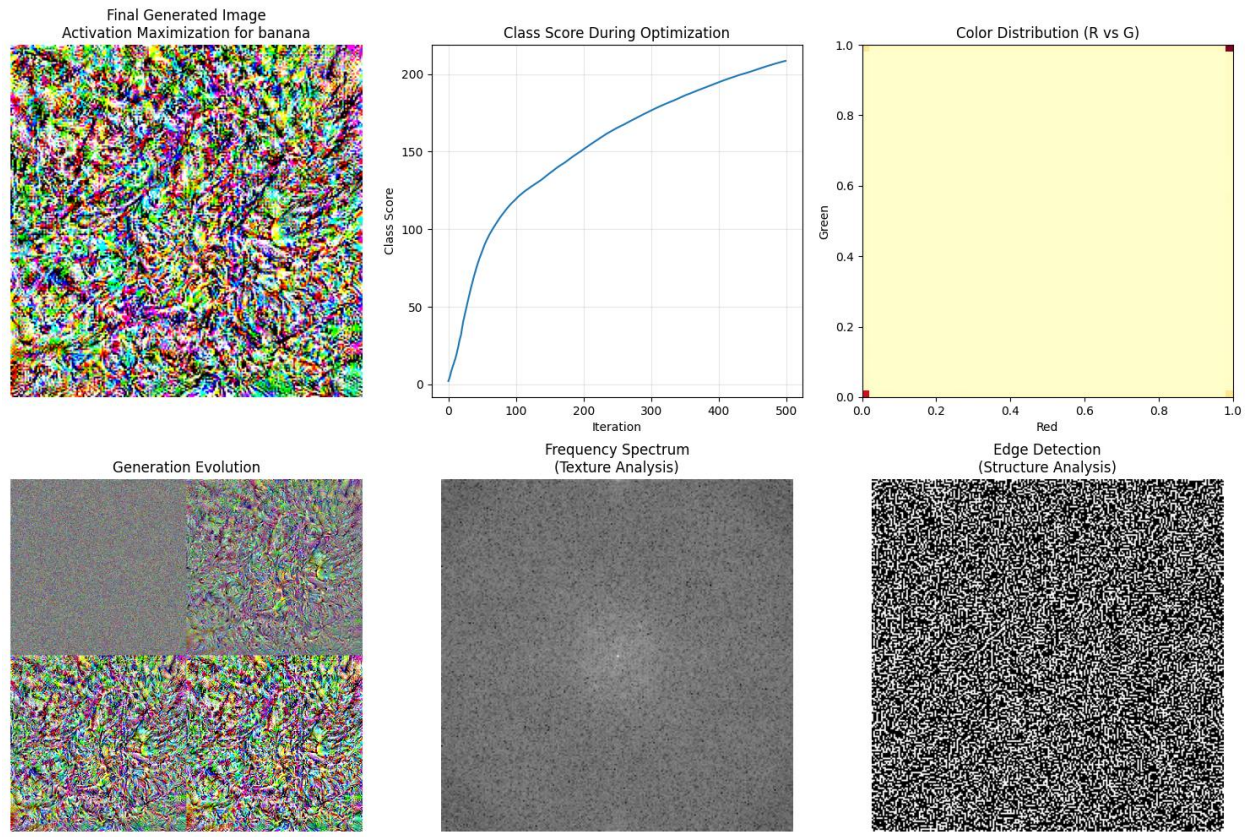
Iteration 100: class score = 119.4459

Iteration 200: class score = 151.5215

Iteration 300: class score = 176.4856

Iteration 400: class score = 194.6883

Final iteration 500: class score = 208.4723



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COMPARING DIFFERENT REGULARIZATION TECHNIQUES

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Generating with: No Regularization

```
Starting activation maximization for class 954
Optimization parameters:
  - Iterations: 300
  - Learning rate: 0.05
  - L2 regularization: 0
  - TV regularization: 0
  - Gaussian blur: False (sigma=0.5)
--------------------------------------------------
Iteration 0: class score = 1.9251
Iteration 100: class score = 113.5043
Iteration 200: class score = 135.8075
Final iteration 300: class score = 154.8168


Generating with: L2 Only
Starting activation maximization for class 954
Optimization parameters:
  - Iterations: 300
  - Learning rate: 0.05
  - L2 regularization: 0.01
  - TV regularization: 0
  - Gaussian blur: False (sigma=0.5)
--------------------------------------------------
Iteration 0: class score = 2.0436
Iteration 100: class score = 112.5255
Iteration 200: class score = 139.6688
Final iteration 300: class score = 161.6865
```

```
Generating with: TV Only
Starting activation maximization for class 954
Optimization parameters:
  - Iterations: 300
  - Learning rate: 0.05
  - L2 regularization: 0
  - TV regularization: 0.01
  - Gaussian blur: False (sigma=0.5)
--------------------------------------------------
Iteration 0: class score = 2.0405
Iteration 100: class score = 121.3268
Iteration 200: class score = 151.4371
Final iteration 300: class score = 176.2171

Generating with: Blur Only
Starting activation maximization for class 954
Optimization parameters:
  - Iterations: 300
  - Learning rate: 0.05
  - L2 regularization: 0
  - TV regularization: 0
  - Gaussian blur: True (sigma=0.5)
--------------------------------------------------
Iteration 0: class score = 1.8989
Iteration 100: class score = 137.4983
```

```
Iteration 200: class score = 172.7923
Final iteration 300: class score = 136.0547

Generating with: L2 + TV
Starting activation maximization for class 954
Optimization parameters:
  - Iterations: 300
  - Learning rate: 0.05
  - L2 regularization: 0.005
  - TV regularization: 0.005
  - Gaussian blur: False (sigma=0.5)
--------------------------------------------------
Iteration 0: class score = 2.0051
Iteration 100: class score = 126.5383
Iteration 200: class score = 156.3727
Final iteration 300: class score = 179.7255

Generating with: L2 + Blur
Starting activation maximization for class 954
Optimization parameters:
  - Iterations: 300
  - Learning rate: 0.05
  - L2 regularization: 0.005
  - TV regularization: 0
  - Gaussian blur: True (sigma=0.5)
--------------------------------------------------
```

Iteration 0: class score = 1.9317

Iteration 100: class score = 146.4411

Iteration 200: class score = 185.0261

Final iteration 300: class score = 150.6976

Generating with: TV + Blur

Starting activation maximization for class 954

Optimization parameters:

- Iterations: 300
- Learning rate: 0.05
- L2 regularization: 0
- TV regularization: 0.005
- Gaussian blur: True (sigma=0.5)

Iteration 0: class score = 2.0535

Iteration 100: class score = 142.2697

Iteration 200: class score = 178.2709

Final iteration 300: class score = 151.0581

Generating with: All Combined

Starting activation maximization for class 954

Optimization parameters:

- Iterations: 300
- Learning rate: 0.05
- L2 regularization: 0.001
- TV regularization: 0.001

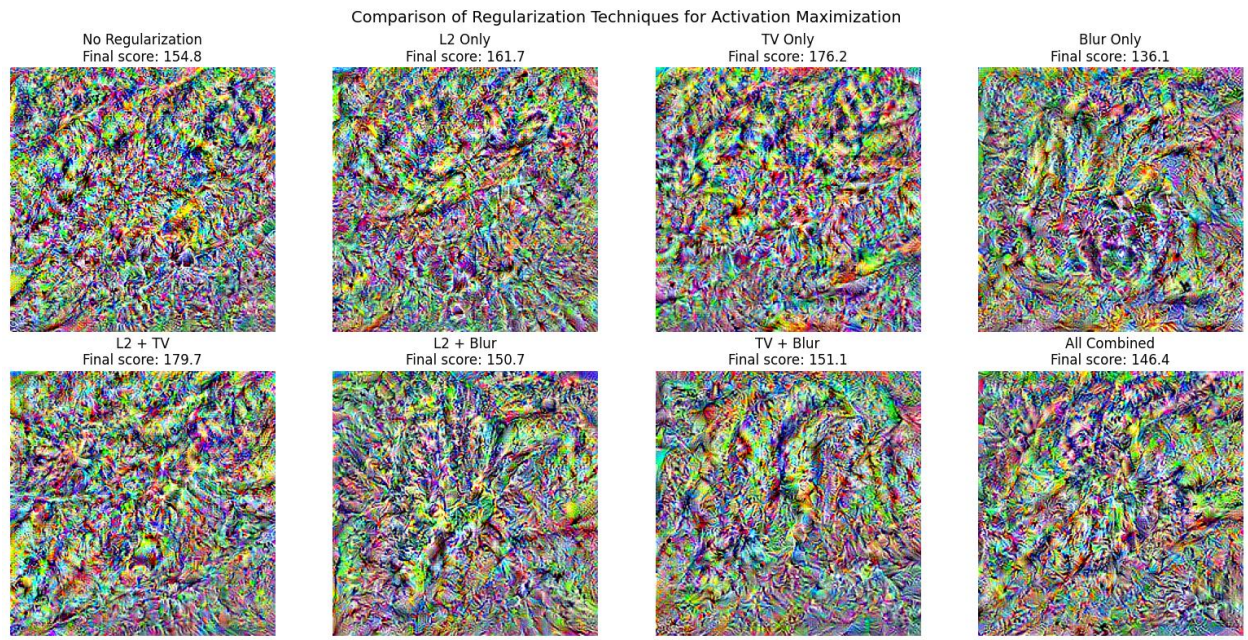
- Gaussian blur: True (sigma=0.5)

Iteration 0: class score = 2.0583

Iteration 100: class score = 144.5407

Iteration 200: class score = 180.2700

Final iteration 300: class score = 146.4430



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EXPECTED INTERPRETATION FOR TASK 2:

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Smooth Visualization via Activation Maximization Analysis:

1. Generated Pattern Characteristics:

   - The optimization produces yellow curved textures that resemble banana features

   - Patterns show elongated curved structures similar to banana shape

   - Yellow/orange color dominance reflects the model's association with banana color

2. Texture Bias of CNN:

- The generated image emphasizes textures over global shape
- Fine-scale patterns and repeated structures are prominent
- This demonstrates that CNNs heavily rely on texture information for classification
- The model learns that bananas have specific textural properties (smooth with spots)

3. Regularization Effects:

- Without regularization: Images become noisy and unrecognizable
- With L2 regularization: Prevents extreme pixel values
- With Total Variation (TV): Encourages smoother transitions between regions
- With Gaussian blur: Enforces spatial coherence
- Best results come from combining multiple regularization techniques

4. Visual Patterns:

- Configuration 1 (Balanced): Shows clear banana-like textures with curved patterns
- Configuration 2 (Aggressive): Produces smoother, more abstract patterns
- Configuration 3 (Minimal): Results in noisy, less interpretable images

5. Model Insights:

- The optimization reveals what features the banana class neuron "looks for"
- Strong activation for yellow-orange colors indicates color importance
- Curved patterns suggest the model recognizes banana shape
- Texture patterns reveal the model's reliance on surface appearance

6. Without Regularization:

- Images may look noisy and unnatural
- High-frequency patterns dominate
- The optimization exploits model weaknesses
- Regularization is essential for interpretable results

Key Takeaway:

Activation maximization shows that the banana class neuron responds strongly to yellow, curved, textured patterns. This reveals the model's texture bias - it learns that bananas are characterized by specific textural properties rather than just shape. Different regularization techniques produce varying levels of interpretability, with combined approaches yielding the most meaningful results.